Mathematical Language

Notation and Conventions

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

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About This Document

Mathematical Language records the notation and writing conventions adopted here. It is guided by deliberate personal preferences together with established mathematical practice; when no personal convention has been fixed, standard mathematical usage is the default.

This document is intentionally non-exhaustive and may be extended as new subjects and notation are encountered. A displayed example or notation-table entry gives a standard or preferred form, not an exhaustive grammar for every related mathematical expression, unless explicitly stated otherwise. When multiple alternative notations for the same object or purpose are listed, they are given in preferred order unless stated otherwise: the primary notation is used by default, while listed secondary notation remains acceptable when the surrounding context, an adopted textbook or lecture note, an established field convention, or the notation of a given problem makes it preferable. The absence of a notation from this document does not by itself make that notation invalid.

When notation not covered here is needed, an established standard should be used when suitable; otherwise, a clear, mathematically coherent, and unambiguous notation appropriate to the context may be introduced. Useful choices may later be incorporated into this document. These conventions should be applied with mathematical judgment rather than mechanically: explicitly stated conventions should be respected, while established field-specific usage, semantic distinctions, and context take precedence over superficial uniformity.

Part I. General Conventions and Writing

1 General Conventions

Parentheses are preferred for ordinary grouping when delimiters are otherwise interchangeable, including in nested expressions; nesting alone is not a reason to alternate mechanically among parentheses, square brackets, and braces. Square brackets, braces, angle brackets, and other delimiters are used whenever they have an established field-specific, structural, or semantic role.

Item	Convention
Scalars	Ordinary mathematical italic letters, for example a, b, c, λ, t.
Vectors	Bold letters; generic vectors use lowercase letters such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$, while established field-specific vector names may use other letter cases. Unless otherwise stated, vectors are column vectors.
Matrices	Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$. Parentheses are the preferred matrix delimiters; square brackets are a standard alternative.
Greek vectors and matrices	Bold Greek letters when a Greek symbol denotes a vector or matrix, for example $\boldsymbol{\xi}$ and $\boldsymbol{\Phi}$.
Dimension subscripts	Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript.
Number systems	Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
Fields	Arbitrary or named fields are denoted by ordinary italic uppercase letters such as F, K, L. Standard finite fields such as $\mathbb{F}_q$ retain blackboard bold; other established blackboard-bold notation follows the standard convention of the relevant field.
Sets	Ordinary uppercase letters such as A, B, X, Y, unless a field-specific convention requires otherwise.
Families and collections	Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$.
Symbol distinctions	Different mathematical letterforms and alphabets are treated as distinct symbols, even when based on the same letter. For example, ε and ϵ, φ and ϕ, and A, $\mathbf{A}$, $\mathcal{A}$, $\mathbb{A}$, and $\mathfrak{A}$ are distinct symbols.

Item	Convention
Named operators	Upright Roman type, for example det, ker, im, rk, and tr.
Large operators	Use display-style sizing and conventional index placement by default throughout mathematical writing, including proofs, solutions, examples, inline expressions, and notation tables. This applies to sums, products, indexed unions and intersections, integrals, limits (including limit superior and inferior), extrema, and analogous operators. Indices for sums, products, and similar operators are placed above and below; those for limits and extrema are placed below; integral bounds remain to the right. Smaller sizes or side-positioned indices are permitted only when clearly justified by readability or deliberate typographic layout, such as in fractions or tightly nested expressions, and not merely for convenience or routine space saving.

Notation	Meaning
$(a_1, \dots, a_n)$	Ordered tuple
$(a_n),\ \{a_n\}$	Sequence
$(a_i)_{i\in I},\ \{a_i\}_{i\in I}$	Indexed family over an arbitrary index set I
(a, b)	Open interval
$[a, b]$	Closed interval
$[a, b)$	Left-closed, right-open interval
$(a, b]$	Left-open, right-closed interval
$\{a_1, \dots, a_n\}$	Set delimiters
$\{x \in A \mid P(x)\}$, $\{x \in A\colon P(x)\}$	Set-builder notation
$\langle x, y\rangle$	Pairing or inner product, according to context
$\lvert x\rvert$	Absolute value, modulus, cardinality, or order, according to context
$\lVert x\rVert$	Norm
$a = b$	Equality
$a \coloneqq b$	a is defined to be b.
$a \equiv b$	Identity, equivalence, or congruence, according to context
$a \cong b$	Isomorphism or congruence, according to context
$a \approx b$	Approximation
$f\colon X \to Y$	Function or map from X to Y
$x \mapsto f(x)$	Elementwise rule of a map
$f\colon X \to Y,\ x \mapsto f(x)$	Function specified by its domain, codomain, and elementwise rule
$g \circ f$	Composition: first f, then g
$f\lvert_A,\ f \restriction_A$	Restriction of f to A
id_X	Identity map on X

2 Mathematical Writing

Mathematical Environments.

Item	Convention
Definition	Precise meaning of a term, object, property, or construction.
Notation	Introduction or declaration of notation.
Convention	Standing choice of notation, terminology, normalization, or usage adopted for the stated context.

Item	Convention
Motivation	Explanation of why a concept, method, problem, or development is introduced or worth considering.
Axiom	Fundamental statement accepted without proof as part of a mathematical system or theory.
Assumption	Condition adopted as a premise for the current statement, argument, construction, or discussion.
Hypothesis	Distinguished condition under which a result or theory is considered; statements conventionally named hypotheses retain that terminology.
Conjecture	Statement proposed to be true but not proved in full generality.
Theorem	Principal mathematical result.
Proposition	Useful result of more limited scope.
Lemma	Supporting result, usually for a later proof.
Corollary	Result following readily from a preceding result.
Claim	Local or intermediate assertion within a larger argument.
Fact	Established statement used without the emphasis or status of a named theorem.
Observation	Mathematical fact or consequence readily noticed or inferred from the immediate context, definitions, or preceding results.
Intuition	Informal or heuristic insight into a mathematical concept, statement, or problem, suggesting what to expect or why a result seems plausible without constituting a proof.
Proof	Rigorous argument establishing a stated mathematical assertion.
Sketch	Outline of the essential reasoning, construction, or solution with some details omitted.
Formula	Distinguished computational identity or closed form.
Algorithm	Explicit step-by-step procedure.
Construction	Explicit specification of how a mathematical object is defined or built to satisfy stated requirements.
Design	Explanation of the roles of individual components and the rationale for structural choices in a mathematical object, function, method, or environment.
Example	Illustration of a definition, construction, phenomenon, or result.
Counterexample	Example disproving a general assertion or showing that a condition is necessary.
Application	Use of a concept, method, or result in a concrete mathematical or applied setting.
Remark	Supplementary mathematical clarification, qualification, or extension of the surrounding material.
Note	Brief contextual or practical information, such as a local reminder, permission, or usage instruction, worth keeping in mind.
Comment	Retrospective expository interpretation or takeaway concerning the significance, lesson, or point of the preceding material.
Warning	Important notice of a subtlety, common error, or potentially misleading point.
Question	Question posed to prompt thought, exploration, or discussion without necessarily requiring a complete solution.
Problem	Mathematical task intended to be solved.
Exercise	Task primarily intended to practice, apply, or reinforce mathematical material.
Open Problem	Problem not known to be solved in the stated form.

Item	Convention
Hint	A partial clue intended to help the reader discover a proof or solution without providing the complete argument or answer.
Suggestion	An optional recommendation to consider a particular observation, method, or approach when solving a problem or exercise.
Solution	Resolution of a stated problem or task by proof, computation, construction, or other appropriate means.
Idea	Key insight guiding a proof, solution, or construction.
Strategy	Broad problem-solving plan.
Tactic	Local step within a larger strategy.

Proof, sketch, and solution labels are written as *Proof*, *Sketch*, and *Solution*, respectively, each followed by a period.

Notation	Meaning
□	Ending for ordinary proofs and solutions
■	Ending for substantial, difficult, or particularly important proofs
Q.E.D.	Ending for exceptionally difficult, significant, or paper-level proofs

Enumeration and Subdivision. Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

Item	Convention
1.1, 1.2, 1.3, ...	Independent child problems or standalone units under a common parent; each is a distinct subproblem that may be stated, solved, and referenced separately.
(1), (2), (3), ...	Ordinary parallel enumeration, such as choices, cases, possibilities, alternatives, examples, or procedural steps; not a subdivision of a single problem or a clause of a formal mathematical statement.
(a), (b), (c), ...	Connected subparts of a single problem or task that share a common statement, setup, or context.
(i), (ii), (iii), ...	Formal mathematical clauses within definitions, theorems, propositions, lemmas, and related statements, such as hypotheses, conditions, properties, conclusions, or equivalent statements.

The distinction between 1.1 and (a) is independence: 1.1 denotes a distinct child problem under a common parent, whereas (a) denotes a connected part of one problem sharing its setup. The distinction between (1) and (i) is function: (1) is used for ordinary parallel enumeration, whereas (i) is reserved for clauses within formal mathematical statements. Nested enumeration preserves these structural roles rather than following a fixed hierarchy of label styles.

Mathematical Shorthand.

Typeset Abbreviations may be used in mathematical documents, textbooks, problem sets, and solutions, as well as in handwritten work, when appropriate to the context and level of formality. Handwritten Shorthand is reserved for personal notes, scratch work, and blackboard writing.

Typeset Abbreviations.

Notation	Meaning
i.e.	that is
e.g.	for example
etc.	and so on
cf.	compare
resp.	respectively
vs.	versus
i.o.w.	in other words
s.t.	such that
iff	if and only if
w.r.t.	with respect to
const	constant
a.e.	almost everywhere
a.s.	almost surely
i.i.d.	independent and identically distributed
N.t.	Note that
w.l.o.g.	without loss of generality
w.t.s.	want to show (that)
e.t.s.	enough to show (that)
t.f.a.e.	the following are equivalent
i.t.s.w.	in the same way
o.t.o.h.	on the other hand
LHS	Left-hand side
RHS	Right-hand side
IVT	Intermediate Value Theorem
EVT	Extreme Value Theorem
MVT	Mean Value Theorem
FTC	Fundamental Theorem of Calculus
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
IVP	Initial Value Problem
BVP	Boundary Value Problem

Handwritten Shorthand.

Notation	Meaning
Def	Definition
Thm	Theorem
Prop	Proposition
Lem	Lemma
Cor	Corollary
Pf	Proof
Sol	Solution
Ex	Example
Exer	Exercise
Rmk	Remark
Sps	Suppose (that)

Notation	Meaning
cont	continuous
cont.ly	continuously
difb	differentiable
difb.ly	differentiably
difl	differential
bdd	bounded
unif	uniform
unif.ly	uniformly
lin	linear
lin.ly	linearly
dep	dependent
indep	independent
approx	approximate
triv	trivial
fnc	function
seq	sequence
eqn	equation
eqlt	equality
ineqlt	inequality
sys	system
comb	combination
vec	vector
mat	matrix
col	column
pt	point
bdy	boundary
nbhd	neighborhood
itvl	interval
op	operation
num	number
val	value
elmt	element
cw	clockwise
ccw	counterclockwise
w/	with
w/o	without

Mathematical Phrases.

Phrase	Use
We find by inspection that ...	Used when a conclusion follows directly from inspection.

Part II. Notation by Subject

3 Set Theory and Logic

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

Notation	Meaning
$\mathbb{N} = \{1, 2, 3, \dots\},\ \mathbb{Z}_{\geq 1},\ \mathbb{Z}^+$	Positive integers
$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\},\ \mathbb{Z}_{\geq 0}$	Nonnegative integers
$\mathbb{Z}$	Integers
$\mathbb{Q}$	Rational numbers
$\mathbb{R}$	Real numbers
$\mathbb{C}$	Complex numbers
$\varnothing,\ \emptyset$	Empty set
$x \in A$	x is an element of A.
$x \notin A$	x is not an element of A.
$A \subseteq B$	A is a subset of B; equality is allowed.
$A \subsetneq B$	A is a proper subset of B.
$A \supseteq B$	A is a superset of B; equality is allowed.
$A \supsetneq B$	A is a proper superset of B.
$A \cup B$	Union
$A \cap B$	Intersection
$(A_i)_{i \in I}$	Indexed family of sets
$\bigcup_{i \in I} A_i$	Union of an indexed family of sets
$\bigcap_{i \in I} A_i$	Intersection of an indexed family of sets
$A \setminus B,\ A - B$	Set difference
$A \triangle B,\ A\, \Delta\, B$	Symmetric difference
A^c	Complement of A in the ambient set
$\mathcal{P}(A),\ 2^A$	Power set
$A \times B$	Cartesian product
$A_1 \times \cdots \times A_n,\ A^n$	Finite Cartesian product; A^n is the n-fold Cartesian power of A
$\prod_{i \in I} A_i$	Cartesian product of an indexed family of sets
$A \sqcup B,\ A \dot\cup B$	Disjoint union
$\bigsqcup_{i \in I} A_i$	Disjoint union of an indexed family of sets
$A \cap B = \varnothing$	A and B are disjoint.
$R \subseteq X \times Y$	Binary relation between X and Y
xRy	x is related to y by the relation R.
R^{-1}	Inverse relation of R
$[x]_R,\ [x]$	Equivalence class of x under R; the relation subscript is omitted when the relation is clear
$x \sim y$	Equivalent elements under an understood equivalence relation
$A/\!\sim$	Set of equivalence classes
$\operatorname{dom} f$	Domain of f
$\operatorname{codom} f$	Codomain of f
$f(A)$	Image of A under f

Notation	Meaning
$f^{-1}(B)$	Preimage of B under f; this notation does not require f to be invertible
$\operatorname{im} f$	Image of the map f
Y^X	Set of all functions from X to Y
$f^{-1}\colon Y \to X$	Inverse function of a bijection $f\colon X \to Y$
$f\colon X \hookrightarrow Y$	Injective map
$f\colon X \twoheadrightarrow Y$	Surjective map
$f\colon X \hookrightarrow\!\!\!\!\twoheadrightarrow Y$	Bijective map
$f\colon X \xrightarrow{\sim} Y$	Isomorphism; for sets, equivalently a bijection
$\mathbb{1}_A(x)$	Indicator function of A
$\forall x \in A$	For every $x \in A$
$\exists x \in A$	There exists $x \in A$.
$\exists! x \in A$	There exists a unique $x \in A$.
$\nexists x \in A$	There does not exist $x \in A$.
$P \Rightarrow Q$	P implies Q in ordinary mathematical writing.
$P \Leftarrow Q$	Q implies P in ordinary mathematical writing.
$P \not\Rightarrow Q$	P does not imply Q.
$P \Leftrightarrow Q$	P if and only if Q in ordinary mathematical writing.
$P \wedge Q$	Logical conjunction
$P \vee Q$	Logical disjunction
$\neg P$	Logical negation
$\top,\ \bot$	Truth and falsehood constants, respectively, in formal logic
$\varphi \to \psi,\ \varphi \leftrightarrow \psi$	Conditional and biconditional connectives in formal logic
$\mathcal{M} \models \varphi,\ \mathcal{M} \not\models \varphi$	$\mathcal{M}$ satisfies, or does not satisfy, the formula φ.
$\Gamma \models \varphi,\ \Gamma \not\models \varphi$	φ is, or is not, a semantic consequence of Γ.
$\Gamma \vdash \varphi,\ \Gamma \nvdash \varphi$	φ is, or is not, syntactically derivable from Γ.
$\lightning$	Contradiction reached in a proof or argument; distinct from the formal falsehood constant $\bot$
$\therefore$	Therefore; indicates a conclusion from preceding statements
$\because$	Because; indicates a reason or justification
$\lvert A\rvert,\ \#A$	Cardinality; $\#A$ is also used when emphasizing the number of elements
$\aleph_0$	Cardinality of a countably infinite set
$\mathfrak{c}$	Cardinality of the continuum

4 Calculus and Analysis

The differential d is upright; ε is the default small positive quantity in analysis. Function spaces take their domains in parentheses, for example $C([a,b])$, $\mathcal{R}([a,b])$, and $L^p(\Omega)$.

Notation	Meaning
$\lim_{x \to a} f(x)$	Limit as x approaches a
$\lim_{x \to a^-} f(x)$	Left-hand limit as x approaches a
$\lim_{x \to a^+} f(x)$	Right-hand limit as x approaches a
$\lim_{x \to \infty} f(x),\ \lim_{x \to -\infty} f(x)$	Limits as x tends to positive or negative infinity

Notation	Meaning
$\lim_{x\to a} f(x) = \infty$, $\lim_{x\to a} f(x) = -\infty$	Infinite limits
$\min A$	Minimum of A
$\max A$	Maximum of A
$\inf A$	Infimum of A
$\sup A$	Supremum of A
$\arg\min_{x\in A} f(x)$	Set of minimizers of f over A; empty if the infimum is not attained
$\arg\max_{x\in A} f(x)$	Set of maximizers of f over A; empty if the supremum is not attained
$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$	Extended real line
$f'(a)$	First derivative of f at a
$f^{(n)}(a)$	nth derivative of f at a
$\frac{\mathrm{d}y}{\mathrm{d}x}$	Derivative of y with respect to x
$\frac{\mathrm{d}^n y}{\mathrm{d}x^n}$	nth derivative of y with respect to x in Leibniz notation
$\frac{\partial f}{\partial x_i}$, $\partial_i f$, f_{x_i}	Partial derivative with respect to x_i
$\frac{\partial^2 f}{\partial x_j \partial x_i}$, $\partial_{ji} f$, $f_{x_i x_j}$	Successive or mixed partial derivative, first with respect to x_i and then x_j. When the relevant mixed partial derivatives are continuous, the order may be interchanged.
$D_{\mathbf{v}} f(\mathbf{a})$, $\partial_{\mathbf{v}} f(\mathbf{a})$	Directional derivative along the vector $\mathbf{v}$, which need not be a unit vector: $D_{\mathbf{v}} f(\mathbf{a}) = \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{a} + t\mathbf{v})\Big\vert_{t=0} = Df(\mathbf{a})[\mathbf{v}] = \nabla f(\mathbf{a}) \cdot \mathbf{v}$ when f is differentiable. For the unit direction determined by $\mathbf{v} \neq \mathbf{0}$, use $\mathbf{u} = \mathbf{v}/\lVert\mathbf{v}\rVert$
$\mathrm{d}f = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\, \mathrm{d}x_i$	Total differential of a scalar function of n variables
$\int f(x)\, \mathrm{d}x$	Indefinite integral; a family of antiderivatives, with an additive constant understood
$\int_a^b f(x)\, \mathrm{d}x$	Definite integral over $[a, b]$
$\int_a^\infty f(x)\, \mathrm{d}x$, $\int_{-\infty}^b f(x)\, \mathrm{d}x$	Improper integrals over unbounded intervals
$\int_a^b f\, \mathrm{d}\alpha$	Riemann–Stieltjes integral of f with respect to α
$P = \{x_0, \ldots, x_n\}$, $a = x_0 < \cdots < x_n = b$	Partition of the interval $[a, b]$
$\sum_{i=1}^{n} f(\xi_i)\Delta x_i$, $\Delta x_i = x_i - x_{i-1}$	Tagged Riemann sum, with $\xi_i \in [x_{i-1}, x_i]$
$\iint_D f\, \mathrm{d}A$	Double integral over D
$\iiint_\Omega f\, \mathrm{d}V$	Triple integral over Ω
$\int_C f(z)\, \mathrm{d}z$	Contour integral along an oriented contour C
$\oint_C f(z)\, \mathrm{d}z$	Contour integral over a closed contour C
$-C$	The contour C with the opposite orientation

Notation	Meaning
γ^{-1}	Reversed parametrized path γ, when unambiguous; not a functional inverse
PV, P.V.	Principal value; in particular, the Cauchy principal value of a singular or improper integral
$\sum_{k=1}^{n} a_k$	Finite sum
$\sum_{k=1}^{\infty} a_k$	Infinite series
$\prod_{k=1}^{n} a_k$	Finite product
$\sum_{n=0}^{\infty} a_n(x-a)^n$	Power series centered at a
$\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$	Laurent series about z_0, on an annulus of convergence
$\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$	nth Taylor polynomial of f about a; no single letter for the polynomial is imposed globally
e^x, $\exp x$	Exponential function
$\ln x$, $\log_e x$	Natural logarithm of a positive real number
$\lg x$, $\log_{10} x$	Logarithm to base 10
$\operatorname{lb} x$, $\log_2 x$	Logarithm to base 2
$\log_a x$	Logarithm to base a
$\sin x$	Sine
$\cos x$	Cosine
$\tan x$	Tangent
$\csc x$	Cosecant
$\sec x$	Secant
$\cot x$	Cotangent
$\arcsin x$, $\sin^{-1} x$	Principal inverse sine, $[-1,1] \to [-\pi/2, \pi/2]$
$\arccos x$, $\cos^{-1} x$	Principal inverse cosine, $[-1,1] \to [0, \pi]$
$\arctan x$, $\tan^{-1} x$	Principal inverse tangent, $\mathbb{R} \to (-\pi/2, \pi/2)$
$\operatorname{arccsc} x$, $\csc^{-1} x$	Principal inverse cosecant, defined by $\operatorname{arccsc} x = \arcsin(1/x)$ for $\lvert x\rvert \geq 1$; its range is $[-\pi/2, 0) \cup (0, \pi/2]$
$\operatorname{arcsec} x$, $\sec^{-1} x$	Principal inverse secant, defined by $\operatorname{arcsec} x = \arccos(1/x)$ for $\lvert x\rvert \geq 1$; its range is $[0, \pi/2) \cup (\pi/2, \pi]$
$\operatorname{arccot} x$, $\cot^{-1} x$	Principal real inverse cotangent, $\mathbb{R} \to (0, \pi)$
$\sinh x$	Hyperbolic sine
$\cosh x$	Hyperbolic cosine
$\tanh x$	Hyperbolic tangent
$\operatorname{csch} x$	Hyperbolic cosecant
$\operatorname{sech} x$	Hyperbolic secant
$\coth x$	Hyperbolic cotangent

Notation	Meaning
$\operatorname{arcsinh} x,\ \sinh^{-1} x$	Principal real inverse hyperbolic sine, $\mathbb{R} \to \mathbb{R}$
$\operatorname{arccosh} x,\ \cosh^{-1} x$	Principal real inverse hyperbolic cosine, $[1, \infty) \to [0, \infty)$
$\operatorname{arctanh} x,\ \tanh^{-1} x$	Principal real inverse hyperbolic tangent, $(-1, 1) \to \mathbb{R}$
$\operatorname{arccsch} x,\ \operatorname{csch}^{-1} x$	Principal real inverse hyperbolic cosecant, defined by $\operatorname{arccsch} x = \operatorname{arcsinh}(1/x)$ on $\mathbb{R} \setminus \{0\}$; its range is $\mathbb{R} \setminus \{0\}$
$\operatorname{arcsech} x,\ \operatorname{sech}^{-1} x$	Principal real inverse hyperbolic secant, defined by $\operatorname{arcsech} x = \operatorname{arccosh}(1/x)$ on $(0, 1]$; its range is $[0, \infty)$
$\operatorname{arccoth} x,\ \coth^{-1} x$	Principal real inverse hyperbolic cotangent, defined by $\operatorname{arccoth} x = \operatorname{arctanh}(1/x)$ for $\lvert x \rvert > 1$; its range is $\mathbb{R} \setminus \{0\}$
$\Gamma(z)$	Gamma function
$B(x, y)$	Beta function
$\zeta(s)$	Riemann zeta function
$\mathbf{J}_{\mathbf{F}}(\mathbf{a}),\ D\mathbf{F}(\mathbf{a})$	Jacobian matrix of $\mathbf{F} = (F_1, \ldots, F_m) \colon \mathbb{R}^n \to \mathbb{R}^m$ at $\mathbf{a}$, with $\mathbf{J}_{\mathbf{F}}(\mathbf{a}) = \left(\dfrac{\partial F_i}{\partial x_j}(\mathbf{a}) \right)_{i,j}$. The notation $D\mathbf{F}(\mathbf{a})$ denotes the total derivative, identified with this matrix in standard coordinates
$\det \mathbf{J}_{\mathbf{F}}(\mathbf{a}),\ \det D\mathbf{F}(\mathbf{a}),$ $\left.\dfrac{\partial(F_1, \ldots, F_n)}{\partial(x_1, \ldots, x_n)}\right\rvert_{\mathbf{a}}$	Jacobian determinant of $\mathbf{F} \colon \mathbb{R}^n \to \mathbb{R}^n$ at $\mathbf{a}$
$\operatorname{Hess} f$	Hessian matrix of f
$\nabla f,\ \operatorname{grad} f$	Gradient of f
$\nabla \cdot \mathbf{F},\ \operatorname{div} \mathbf{F}$	Divergence of $\mathbf{F}$
$\nabla \times \mathbf{F},\ \operatorname{curl} \mathbf{F}$	Curl of $\mathbf{F}$
Δf	Laplacian of f
$\mathbf{r}(t)$	Parametrized curve
$\mathrm{d}s$	Element of arc length
$\displaystyle\int_C f \, \mathrm{d}s$	Scalar line integral along C with respect to arc length
$\displaystyle\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	Line integral of the vector field $\mathbf{F}$ along C
$\displaystyle\iint_S f \, \mathrm{d}S$	Scalar surface integral over S
$\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S$	Flux integral through S
$x_n \to x,\ \lim_{n\to\infty} x_n = x$	Convergence of a sequence
(x_{n_k})	Subsequence of the sequence (x_n)
$a_n \nearrow L$	Monotone increasing convergence to L
$a_n \searrow L$	Monotone decreasing convergence to L
$\limsup_{n\to\infty} a_n$	Limit superior of the sequence (a_n)
$\liminf_{n\to\infty} a_n$	Limit inferior of the sequence (a_n)
$f(x) \sim g(x)$	Asymptotic equivalence in the stated limiting regime, typically meaning $f(x)/g(x) \to 1$
$f_n \to f,\ f_n \xrightarrow{\text{ptw}} f$	Pointwise convergence of functions on a common domain by default when no ambient topology, metric, or norm is specified. If such a structure is specified, the ordinary arrow $\to$ refers to convergence in that structure
$f_n \rightrightarrows f,\ f_n \xrightarrow{\text{unif}} f$	Uniform convergence
$f_n \xrightarrow{\text{a.e.}} f$	Convergence almost everywhere
$(X, \mathcal{A}, \mu)$	Measure space, where $\mathcal{A}$ is a σ-algebra on X and μ is a measure on $\mathcal{A}$

Notation	Meaning
$\mathcal{A}$	σ-algebra of measurable subsets of X, when the underlying set is understood
$\mathcal{B}(X)$	Borel σ-algebra of a topological space X
$\mu(E)$	Measure of a measurable set $E \in \mathcal{A}$
$\displaystyle\int_E f \, \mathrm{d}\mu$	Integral of f over E with respect to the measure μ
$f = g$ a.e., $\quad f = g$ μ-a.e.	Equality almost everywhere, with the measure omitted when understood
$\operatorname{ess\,sup}_{x \in E} f(x)$	Essential supremum of f on E
$\operatorname{ess\,inf}_{x \in E} f(x)$	Essential infimum of f on E
$C(\Omega),\ C^0(\Omega)$	Continuous functions on Ω
$C^k(\Omega)$	k times continuously differentiable functions on Ω
$C^\infty(\Omega)$	Smooth functions on Ω
$\operatorname{supp} f$	Support of f, the closure of the set on which f is nonzero
$C_c(\Omega),\ C_c^\infty(\Omega)$	Continuous and smooth functions on Ω with compact support, respectively
$\mathcal{R}([a,b])$	Riemann-integrable functions on $[a,b]$
$L^p(\Omega),\ L^p(X,\mu)$	Lebesgue L^p spaces; the measure is displayed when it is not understood
$L^\infty(\Omega),\ L^\infty(X,\mu)$	Lebesgue spaces of essentially bounded functions
ℓ^p	Space of scalar sequences (x_n) with $\displaystyle\sum_{n=1}^{\infty} \lvert x_n\rvert^p < \infty$, for $1 \le p < \infty$
ℓ^∞	Space of bounded scalar sequences
$\lVert f\rVert_\infty$	Supremum norm $\sup_{\mathbf{x} \in \Omega} \lvert f(\mathbf{x})\rvert$ for an actual bounded function on a Euclidean domain, especially in bounded- or continuous-function spaces
$\lVert f\rVert_{L^p(X,\mu)}$	L^p-norm $\left(\displaystyle\int_X \lvert f\rvert^p \, \mathrm{d}\mu\right)^{1/p}$ for $1 \le p < \infty$ on a measure space $(X, \mathcal{A}, \mu)$
$\lVert f\rVert_{L^p(\Omega)},\ \lVert f\rVert_p$	L^p-norm $\left(\displaystyle\int_\Omega \lvert f(\mathbf{x})\rvert^p \, \mathrm{d}\mathbf{x}\right)^{1/p}$ for $1 \le p < \infty$ on a Euclidean domain when the underlying measure is understood
$\lVert f\rVert_{L^\infty(\Omega)}$	L^∞-norm $\operatorname{ess\,sup}_{\mathbf{x} \in \Omega} \lvert f(\mathbf{x})\rvert$ on a Euclidean domain; distinguished from the ordinary supremum norm
$\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$	Multi-index; α is not treated as a vector for typographic purposes
$\lvert\alpha\rvert = \alpha_1 + \cdots + \alpha_n$, $\alpha! = \alpha_1! \cdots \alpha_n!$	Order and factorial of a multi-index
$\mathbf{x}^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$	Multi-index monomial for $\mathbf{x} = (x_1, \ldots, x_n)$
$D^\alpha f,\ \partial^\alpha f$	Mixed partial derivative specified by the multi-index α
$B_r(x)$	Open ball of radius r centered at x
$\overline{B}_r(x) = \{y \mid d(x,y) \le r\}$	Closed ball of radius r centered at x in a metric space (X,d). The bar is part of the closed-ball notation; in an arbitrary metric space, $\overline{B}_r(x)$ need not equal the closure $\overline{B_r(x)}$ of the open ball
$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$	Extended complex plane, equivalently the Riemann sphere
$i^2 = -1$	Imaginary unit
$z = x + iy$	Complex number with real part x and imaginary part y
$\operatorname{Re} z,\ \Re z$	Real part of z
$\operatorname{Im} z,\ \Im z$	Imaginary part of z; distinct from lowercase im for image

Notation	Meaning
$\overline{z}$	Complex conjugate
$\lvert z\rvert$	Complex modulus
$\arg z$	Argument of a nonzero complex number z; its values differ by integer multiples of 2π
$\operatorname{Arg} z$	Principal argument of $z \neq 0$, with $-\pi < \operatorname{Arg} z \leq \pi$
$\log z,\ \ln z$	General multivalued complex logarithm for $z \neq 0$; $\log z = \{\ln\lvert z\rvert + i(\operatorname{Arg} z + 2\pi k) \mid k \in \mathbb{Z}\}$
$\operatorname{Log} z,\ \operatorname{Ln} z$	Principal complex logarithm for $z \neq 0$; $\operatorname{Log} z = \ln\lvert z\rvert + i \operatorname{Arg} z$. On $\mathbb{C} \setminus (-\infty, 0]$, it is the analytic principal branch, with $-\pi < \operatorname{Arg} z < \pi$
$\operatorname{Res}(f; a),\ \operatorname{Res}_{z=a} f(z)$	Residue of f at a finite point a
$\operatorname{Res}(f; \infty),\ \operatorname{Res}_{z=\infty} f(z)$	Residue of f at infinity

5 Linear Algebra

Notation	Meaning
$M_{m\times n}(F),\ F^{m\times n}$	Space of $m \times n$ matrices over the field F
$M_n(F)$	Space (and matrix algebra) of $n \times n$ matrices over F
$\mathbf{A} = (a_{ij}),\ \mathbf{A} = [a_{ij}]$	Matrix with (i, j)-entry a_{ij}
$\mathbf{A}^{\mathsf{T}}$	Transpose of $\mathbf{A}$
$\mathbf{A}^{\mathsf{H}}$	Conjugate transpose of $\mathbf{A}$
$\mathbf{A}^{-1}$	Inverse of $\mathbf{A}$
$\mathbf{I}_n,\ \mathbf{I}$	Identity matrix; the dimension may be omitted when clear
$\mathbf{0}$	Zero vector or zero matrix, with dimension understood
$\mathbf{e}_i$	ith standard basis vector
$\mathbf{1}$	All-ones vector
$\mathbf{D}$	Diagonal matrix when a local name is required
$\operatorname{diag}(d_1, \ldots, d_n)$	Diagonal matrix with entries $d_1, \ldots, d_n$
$\mathbf{P}$	Projection matrix when a local name is required
$\det(\mathbf{A}),\ \lvert\mathbf{A}\rvert$	Determinant
$\operatorname{rk}(\mathbf{A}),\ \operatorname{rank}(\mathbf{A})$	Rank
$\operatorname{tr}(\mathbf{A})$	Trace
$\operatorname{adj}(\mathbf{A})$	Adjugate matrix
M_{ij}	Minor of the (i, j)-entry, when the matrix is understood
$C_{ij} = (-1)^{i+j} M_{ij}$	Cofactor of the (i, j)-entry
$\operatorname{Col}(\mathbf{A})$	Column space
$\operatorname{Row}(\mathbf{A})$	Row space
$\ker \mathbf{A},\ \operatorname{Null}(\mathbf{A})$	Null space of $\mathbf{A}$; $\operatorname{Null}(\mathbf{A})$ is a matrix-oriented alternative
V, W	Vector spaces, usually over an understood field F
$\operatorname{Hom}_F(V, W)$	Vector space of F-linear maps from V to W
$\operatorname{End}_F(V)$	F-algebra of F-linear endomorphisms of V
$\dim V$	Dimension of V
$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$	Span of the listed vectors
$U \oplus V$	Direct sum
$\ker T$	Kernel of a linear map T
$\operatorname{nullity}(T) = \dim \ker T$	Nullity of a linear map T
$\operatorname{im} T$	Image of a linear map T

Notation	Meaning
$\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$	Ordered basis
$[\mathbf{v}]_{\mathcal{B}}$	Coordinate vector of $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$
$[T]_{\mathcal{C} \leftarrow \mathcal{B}}$	Matrix of $T\colon V \to W$ with respect to domain basis $\mathcal{B}$ and codomain basis $\mathcal{C}$; thus $[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$
$[T]_{\mathcal{B}}$	Matrix of an endomorphism $T\colon V \to V$ when the same basis $\mathcal{B}$ is used in the domain and codomain
$\mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}$	Change-of-coordinate matrix from $\mathcal{B}$-coordinates to $\mathcal{C}$-coordinates; $[\mathbf{v}]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$
V^*	Dual space of V
$\langle \mathbf{u}, \mathbf{v} \rangle,\ (\mathbf{u}, \mathbf{v})$	Inner product. Over complex spaces, the inner product is conjugate-linear in the first argument and linear in the second
$\mathbf{u} \cdot \mathbf{v},\ \mathbf{u}^{\mathsf{T}}\mathbf{v}$	Standard Euclidean inner product on $\mathbb{R}^n$
$\mathbf{u}^{\mathsf{H}}\mathbf{v}$	Standard Hermitian inner product on $\mathbb{C}^n$
$\lVert \mathbf{v} \rVert$	Norm of $\mathbf{v}$
$\mathbf{u} \perp \mathbf{v}$	Orthogonality
$U^{\perp}$	Orthogonal complement of the subspace U
$\operatorname{proj}_U \mathbf{x}$	Orthogonal projection of $\mathbf{x}$ onto the subspace U
$\operatorname{proj}_{\mathbf{v}} \mathbf{x}$	Orthogonal projection of $\mathbf{x}$ onto $\operatorname{span}(\mathbf{v})$
$\mathbf{u} \times \mathbf{v}$	Cross product in $\mathbb{R}^3$
$\mathbf{A}^{\mathsf{T}} = \mathbf{A}$	Real symmetric matrix
$\mathbf{A}^{\mathsf{T}} = -\mathbf{A}$	Real skew-symmetric matrix
$\mathbf{A}^{\mathsf{H}} = \mathbf{A}$	Hermitian matrix
$\mathbf{A}^{\mathsf{H}} = -\mathbf{A}$	Skew-Hermitian matrix
$\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$	Real orthogonal matrix $\mathbf{Q}$
$\mathbf{U}^{\mathsf{H}}\mathbf{U} = \mathbf{I}$	Unitary matrix $\mathbf{U}$
$\mathbf{A}\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\mathsf{H}}\mathbf{A}$	Normal matrix
$\mathbf{P}^2 = \mathbf{P}$	Projection matrix
$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^{\mathsf{T}}$	Orthogonal projection matrix over $\mathbb{R}$
$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^{\mathsf{H}}$	Orthogonal projection matrix over $\mathbb{C}$
$\chi_{\mathbf{A}}(t),\ p_{\mathbf{A}}(t)$	Characteristic polynomial of $\mathbf{A}$, with $\chi_{\mathbf{A}}(t) = \det(t\mathbf{I} - \mathbf{A})$
$m_{\mathbf{A}}(t),\ \mu_{\mathbf{A}}(t)$	Minimal polynomial of $\mathbf{A}$
$\lambda,\ \lambda_i$	Eigenvalue of $\mathbf{A}$
$\mathbf{v},\ \mathbf{v}_i$	Eigenvector of $\mathbf{A}$
$E_\lambda(\mathbf{A}),\ V_\lambda(\mathbf{A})$	Eigenspace associated with λ; $E_\lambda(\mathbf{A}) = \ker(\mathbf{A} - \lambda\mathbf{I})$
$\mathbf{J}_k(\lambda)$	Jordan block of size k with eigenvalue λ, having λ on the diagonal and ones on the superdiagonal
$\sigma(\mathbf{A}),\ \operatorname{spec}(\mathbf{A})$	Spectrum of $\mathbf{A}$
$\rho(\mathbf{A})$	Spectral radius
$\mathbf{P}_\lambda$	Spectral projection associated with the eigenvalue λ, when defined
$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$	Diagonalization of a diagonalizable matrix
$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$	Orthogonal diagonalization of a real symmetric matrix, with $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$
$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{H}}$	Unitary diagonalization of a complex normal matrix, with $\mathbf{U}^{\mathsf{H}}\mathbf{U} = \mathbf{I}$
$\mathbf{A} = \sum_{\lambda \in \sigma(\mathbf{A})} \lambda \mathbf{P}_\lambda$	Spectral decomposition of a diagonalizable matrix, with the sum over distinct eigenvalues
$e^{\mathbf{A}},\ \exp(\mathbf{A})$	Matrix exponential; $e^{t\mathbf{A}}$ is the standard time-dependent form
$\mathbf{A} = \mathbf{L}\mathbf{U}$	LU factorization, with $\mathbf{L}$ lower triangular and $\mathbf{U}$ upper triangular

Notation	Meaning
$\mathbf{A} = \mathbf{QR}$	QR factorization, with $\mathbf{Q}$ orthogonal or unitary and $\mathbf{R}$ upper triangular
$\mathbf{A} = \mathbf{U\Sigma V}^{\mathsf{H}}$	Singular value decomposition (SVD); over $\mathbb{R}$, $\mathbf{V}^{\mathsf{H}} = \mathbf{V}^{\mathsf{T}}$
$\sigma_i(\mathbf{A})$	ith singular value of $\mathbf{A}$
$\mathbf{A}^{\dagger}$	Moore–Penrose pseudoinverse of $\mathbf{A}$
$\mathbf{A} \succ 0$	Positive definite Hermitian matrix (real symmetric in the real case)
$\mathbf{A} \succeq 0$	Positive semidefinite Hermitian matrix
$\mathbf{A} \prec 0$	Negative definite Hermitian matrix
$\mathbf{A} \preceq 0$	Negative semidefinite Hermitian matrix
$\mathbf{A} \succeq \mathbf{B}$	Löwner order: $\mathbf{A} - \mathbf{B} \succeq 0$, for Hermitian matrices of the same size
$\mathbf{A} = \mathbf{LL}^{\mathsf{H}}$	Cholesky factorization of a positive definite Hermitian matrix; over $\mathbb{R}$, $\mathbf{L}^{\mathsf{H}} = \mathbf{L}^{\mathsf{T}}$

6 Differential Equations

Notation	Meaning
y'	First derivative with respect to the understood independent variable
y''	Second derivative
$\dot{x}$	First derivative of x with respect to time
$\ddot{x}$	Second derivative of x with respect to time
$F(t, y, y', \ldots, y^{(n)}) = 0$	General nth-order ordinary differential equation
$y^{(n)} = f(t, y, \ldots, y^{(n-1)})$	nth-order ordinary differential equation in normal form
$y(t_0) = y_0$	Initial condition for a scalar differential equation
$y' + p(t)y = g(t)$	Standard form of a first-order linear differential equation
$\mu(t)$	Integrating factor; for $y' + p(t)y = g(t)$, one may take $\mu(t) = \exp\left(\int p(t)\ \mathrm{d}t\right)$ up to a nonzero constant factor
$M(x,y)\ \mathrm{d}x + N(x,y)\ \mathrm{d}y = 0$	Differential form of a first-order differential equation
M_x, M_y, N_x, N_y	Subscript notation for the corresponding partial derivatives
$L[y]$	Linear differential operator applied to y; $\mathcal{L}$ is reserved here for the Laplace transform
$L[y] = \sum_{k=0}^{n} a_k(t) y^{(k)}$	General nth-order linear differential operator, with $a_n \not\equiv 0$
$L[y] = 0,\ L[y] = g$	Homogeneous and nonhomogeneous linear differential equations, respectively
$y_h,\ y_c$	Homogeneous or complementary part of a solution
y_p	Particular solution
$c_1, c_2, \ldots$	Arbitrary constants in a general solution
$\sum_{k=0}^{n} a_k r^k = 0$	Characteristic equation for a constant-coefficient homogeneous linear differential equation
$r_1, \ldots, r_n$	Characteristic roots; multiplicities are counted when relevant
$W(y_1, \ldots, y_n)(t)$	Wronskian of $y_1, \ldots, y_n$
$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$	Power-series form about the ordinary point x_0
$y = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}$	Frobenius-series form about a regular singular point x_0; r denotes an indicial exponent

Notation	Meaning
$\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$	General first-order system
$\mathbf{x}(t_0) = \mathbf{x}_0$	Initial condition for a first-order system
$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$	Linear first-order system in state-vector form
$\mathbf{x}' = \mathbf{f}(\mathbf{x})$	Autonomous first-order system
$\mathbf{x}_e$	Equilibrium point; $\mathbf{f}(\mathbf{x}_e) = \mathbf{0}$
$\mathbf{\Phi}(t)$	Fundamental matrix of a linear system
$\mathcal{L}\{f(t)\}(s)$, $\mathcal{L}[f](s)$	Laplace transform of f
$\mathcal{L}^{-1}\{F(s)\}(t)$, $\mathcal{L}^{-1}[F](t)$	Inverse Laplace transform of F
$(f * g)(t)$	One-sided convolution, $(f * g)(t) \coloneqq \int_0^t f(\tau) g(t-\tau)\ \mathrm{d}\tau$
$H(t-a),\ u(t-a)$	Heaviside or unit-step function shifted to $t = a$; its value at the jump is specified when relevant
$\delta(t),\ \delta(t-a)$	Dirac delta distribution, or unit impulse, centered at 0 or shifted to $t = a$
$t_n = t_0 + nh$	Uniform numerical grid with step size $h > 0$
$y_n \approx y(t_n)$	Numerical approximation to the exact solution at t_n
$u_x, u_t, u_{xx}, u_{xy}, u_{tt}, \ldots$	Subscript notation for partial derivatives of a scalar function u
$\partial_{\mathbf{n}} u$	Normal derivative, $\partial_{\mathbf{n}} u \coloneqq \nabla u \cdot \mathbf{n}$, for a unit normal vector $\mathbf{n}$
$u\vert_{\partial\Omega} = g$	Dirichlet boundary condition on $\partial\Omega$
$\partial_{\mathbf{n}} u\vert_{\partial\Omega} = g$	Neumann boundary condition on $\partial\Omega$
$\alpha u + \beta\, \partial_{\mathbf{n}} u = g$	Robin boundary condition on $\partial\Omega$
$a_n,\ b_n$	Fourier coefficients for the real $2L$-periodic convention $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$
a_n	$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\ \mathrm{d}x$ for $n \geq 0$
b_n	$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\ \mathrm{d}x$ for $n \geq 1$
$\widehat{f}(\boldsymbol{\xi}),\ \mathcal{F}f(\boldsymbol{\xi})$	Fourier transform under the normalization $\widehat{f}(\boldsymbol{\xi}) \coloneqq \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\ \mathrm{d}\mathbf{x}$; inversion uses $f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\ \mathrm{d}\boldsymbol{\xi}$ whenever applicable
$-(py')' + qy = \lambda w y$	Sturm–Liouville eigenvalue equation, where p, q, and w are functions of x
$\lambda_n,\ \phi_n$	Sturm–Liouville eigenvalue and corresponding eigenfunction
$\langle f, g \rangle_w$	Weighted inner product, $\langle f, g \rangle_w \coloneqq \int_a^b \overline{f(x)} g(x) w(x)\ \mathrm{d}x$
$\langle \phi_m, \phi_n \rangle_w = \delta_{mn}$	Weighted orthonormality of eigenfunctions
$G(x, \xi),\ G(\mathbf{x}, \boldsymbol{\xi})$	Green's function; arguments are scalar or vector as appropriate
$\mathrm{Ai}(x),\ \mathrm{Bi}(x)$	Airy functions
$J_\nu(x)$	Bessel function of the first kind of order ν
$Y_\nu(x),\ N_\nu(x)$	Bessel function of the second kind of order ν
$I_\nu(x),\ K_\nu(x)$	Modified Bessel functions of the first and second kinds
$H_\nu^{(1)}(x),\ H_\nu^{(2)}(x)$	Hankel functions of the first and second kinds
$P_n(x)$	Legendre polynomial of degree n
$P_\nu(x),\ Q_\nu(x)$	Legendre functions of the first and second kinds

Notation	Meaning
$P_\ell^m(x)$	Associated Legendre function for integer degree ℓ and order m
$H_n(x),\ \mathrm{He}_n(x)$	Physicists' and probabilists' Hermite polynomials, respectively
$L_n^{(\alpha)}(x),\ L_n(x)$	Generalized and ordinary Laguerre polynomials; $L_n(x) = L_n^{(0)}(x)$
$T_n(x),\ U_n(x)$	Chebyshev polynomials of the first and second kinds
$P_n^{(\alpha,\beta)}(x)$	Jacobi polynomial
$C_n^{(\lambda)}(x)$	Gegenbauer (ultraspherical) polynomial
$\mathrm{erf}(x),\ \mathrm{erfc}(x)$	Error function and complementary error function
${}_2F_1(a,b;c;z)$	Gauss hypergeometric function

7 Combinatorics and Discrete Mathematics

Notation	Meaning
$[n] = \{1, 2, \ldots, n\}$	Standard n-element set, especially in combinatorics; $[0] = \varnothing$ when needed
$n!$	Factorial
$n!!$	Double factorial: product of positive integers of the same parity not exceeding n, with $0!! = (-1)!! = 1$
$n^{\underline{k}}$	Falling factorial, $n(n-1)\cdots(n-k+1)$
$n^{\overline{k}}$	Rising factorial, $n(n+1)\cdots(n+k-1)$
$(a)_k$	Rising Pochhammer symbol $a(a+1)\cdots(a+k-1)$ in special-functions contexts; it is not used here for the falling factorial
${}_nP_r$	Number of ordered selections of r distinct elements from an n-element set, ${}_nP_r = \dfrac{n!}{(n-r)!}$ for $0 \le r \le n$
$\binom{n}{r},\ {}_nC_r$	Number of r-element subsets of an n-element set
$\left(\!\binom{n}{r}\!\right),\ {}_nH_r$	r-combinations with repetition from n types; equals $\binom{n+r-1}{r}$
$\binom{n}{n_1, \ldots, n_k}$	Multinomial coefficient, for nonnegative integers $n_1, \ldots, n_k$ with $n_1 + \cdots + n_k = n$
$\begin{bmatrix} n \\ k \end{bmatrix},\ c(n,k)$	Unsigned Stirling number of the first kind; permutations of n elements with k cycles
$s(n,k)$	Signed Stirling number of the first kind; $s(n,k) = (-1)^{n-k}c(n,k)$
$\begin{Bmatrix} n \\ k \end{Bmatrix},\ S(n,k)$	Stirling number of the second kind; partitions of an n-element set into k nonempty blocks
$D_n,\ !n$	Number of derangements of n elements; permutations with no fixed points
C_n	nth Catalan number, $C_n = \dfrac{1}{n+1}\binom{2n}{n}$ for $n \ge 0$
B_n	nth Bell number; number of partitions of an n-element set
$p(n)$	Number of integer partitions of n, with $p(0) = 1$
$H_n = \sum_{k=1}^{n} \dfrac{1}{k}$	nth harmonic number for $n \ge 1$, with $H_0 = 0$
$(X, \preceq)$	Partially ordered set (poset)
$\prec,\ \succeq,\ \succ$	Strict and reverse order relations associated with $\preceq$
$\le_{\mathrm{lex}}$	Lexicographic order
$x \lessdot y$	Cover relation: y covers x

Notation	Meaning
B_n	Boolean lattice (Boolean poset) on an n-element ground set
$G = (V, E)$	Graph with vertex set V and edge set E
$V(G)$	Vertex set of G
$E(G)$	Edge set of G
$N(v),\ \Gamma(v)$	Neighborhood of vertex v
$N[v] = N(v) \cup \{v\}$	Closed neighborhood of the vertex v
$\deg_G(v),\ \deg(v)$	Degree of vertex v; the graph subscript may be omitted when clear
$\deg^+(v),\ \deg^-(v)$	Out-degree and in-degree of v in a directed graph
$\delta(G),\ \Delta(G)$	Minimum and maximum vertex degrees, respectively, of a nonempty finite graph G
$d_G(u, v),\ d(u, v)$	Graph distance between vertices u and v
$G[S]$	Subgraph of G induced by the vertex set $S \subseteq V(G)$
$G - v,\ G - S$	Graph obtained by deleting the vertex v, or all vertices in S, together with their incident edges
$G - e$	Graph obtained by deleting the edge e
G/e	Graph obtained by contracting the edge e
$\overline{G}$	Complement of the graph G
K_n	Complete graph on n vertices
$K_{m,n}$	Complete bipartite graph with parts of sizes m and n
P_n	Path graph on n vertices
C_n	Cycle graph on n vertices
$\mathbf{A}(G),\ \mathbf{A}_G$	Adjacency matrix of G
$\mathbf{D}(G),\ \mathbf{D}_G$	Degree matrix of G, the diagonal matrix of vertex degrees in the chosen vertex order
$\mathbf{L}(G),\ \mathbf{L}_G$	Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$
$\chi(G)$	Chromatic number of G
$\alpha(G)$	Independence number of G: maximum size of an independent vertex set
$\omega(G)$	Clique number of G: maximum size of a complete vertex set
$\mathrm{Aut}(G)$	Automorphism group of the graph G
$\tau(G),\ T(G)$	Number of spanning trees of G
$R(k, \ell),\ r(k, \ell),$ $R(k),\ r(k)$	Ramsey numbers; $R(k) = R(k, k)$ in the diagonal case
$A(x) = \sum_{n=0}^{\infty} a_n x^n$	Ordinary generating function of the sequence $(a_n)_{n \geq 0}$
$\sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$	Exponential generating function of the sequence $(a_n)_{n \geq 0}$; the function name is chosen locally when needed
$[x^n]F(x)$	Coefficient of x^n in the formal power series or generating function $F(x)$

8 Probability and Statistics

Probability and expectation are written in blackboard bold by default. Probabilities use parentheses and expectations square brackets by default. Parameter symbols are local to the relevant context: for gamma distributions, β denotes a rate parameter and θ a scale parameter, so $\theta = 1/\beta$, while in hypothesis testing β denotes a Type II error probability. The geometric distribution counts the number of trials up to and including the first success.

Notation	Meaning
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space, with sample space Ω, event σ-algebra $\mathcal{F}$, and probability measure $\mathbb{P}$
$\omega \in \Omega$	Outcome or sample point
$A \in \mathcal{F}$	Event
$X : \Omega \to \mathbb{R}$	Real-valued random variable; measurability from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is understood
$\mathbb{P}(A)$	Probability of event A
$\mathbb{P}(A \mid B)$	Conditional probability of A given B
$X \perp\!\!\!\perp Y$	X and Y are independent.
$X \perp\!\!\!\perp Y \mid Z$	X and Y are conditionally independent given Z.
$p_X(x) = \mathbb{P}(X = x)$	Probability mass function (pmf) of a discrete random variable X
$f_X(x)$	Probability density function (pdf) of a continuous random variable X
$F_X(x) = \mathbb{P}(X \le x)$	Cumulative distribution function (cdf) of X
$p_{X,Y}(x, y),\ f_{X,Y}(x, y)$	Joint pmf or joint pdf of (X, Y), according to context
$p_{X\mid Y}(x \mid y),\ f_{X\mid Y}(x \mid y)$	Conditional pmf or conditional pdf of X given $Y = y$, according to context
$F^{-1}(p)$	Generalized inverse of a cdf F, $F^{-1}(p) = \inf\{x \in \mathbb{R} \mid F(x) \ge p\}$; the p-quantile, for $0 < p < 1$
$\mathbb{E}[X]$	Expectation of random variable X
$\mathbb{E}[X \mid Y]$	Conditional expectation of X given Y
$\mathbb{E}[X \mid \mathcal{G}]$	Conditional expectation of X with respect to a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$
$\mathrm{Var}(X)$	Variance of X
$\sigma_X = \sqrt{\mathrm{Var}(X)}$	Population standard deviation of X
$\mathrm{Cov}(X, Y)$	Covariance of X and Y
$\rho_{X,Y},\ \mathrm{Corr}(X, Y)$	Pearson correlation coefficient of X and Y when both have finite positive variances
r	Sample correlation coefficient, when the sample is understood
$\varphi_X(t) = \mathbb{E}[e^{itX}]$	Characteristic function of X
$M_X(t) = \mathbb{E}[e^{tX}]$	Moment-generating function (mgf) of X, when finite in a neighborhood of 0
$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$	Sample mean as a statistic of a random sample $X_1, \ldots, X_n$
$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	Observed sample mean for data $x_1, \ldots, x_n$
$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$	Sample variance as a statistic
$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	Observed sample variance
$S = \sqrt{S^2},\ s = \sqrt{s^2}$	Sample standard deviation as a statistic and its observed value
$\hat{\theta}$	Estimator or estimate of a parameter θ, according to context
$\hat{p}$	Estimator or estimate of a probability parameter p; commonly the sample proportion
$\mathrm{SE}(\hat{\theta})$	Standard error of an estimator $\hat{\theta}$
$L(\theta; \mathbf{x})$	Likelihood function for parameter θ given observed data $\mathbf{x}$
$\ell(\theta; \mathbf{x}) = \ln L(\theta; \mathbf{x})$	Log-likelihood function
H_0	Null hypothesis
$H_1,\ H_a$	Alternative hypothesis

Notation	Meaning
α	Significance level; an upper bound on the Type I error probability under the specified test
β	Type II error probability under a specified alternative
ν	Degrees of freedom
$X \sim \mathrm{Bern}(p)$	Bernoulli distribution with success probability p
$X \sim \mathrm{Bin}(n,p)$	Binomial distribution
$X \sim \mathrm{Hypergeom}(N,K,n)$	Hypergeometric distribution: number of successes in n draws without replacement from N items, of which K are successes, where $0 \le K, n \le N$. For $\max(0, n-N+K) \le k \le \min(K,n)$, $\mathbb{P}(X=k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$.
$X \sim \mathrm{Geom}(p)$, $\mathbb{P}(X=k) = (1-p)^{k-1}p$	Geometric distribution on $\mathbb{N}$, counting trials up to and including the first success
$X \sim \mathrm{Pois}(\lambda)$	Poisson distribution with parameter λ
$X \sim \mathrm{Unif}(a,b)$	Continuous uniform distribution on $[a,b]$
$X \sim \mathrm{Exp}(\lambda)$	Exponential distribution with rate λ
$X \sim \Gamma(\alpha,\beta)$	Gamma distribution with shape $\alpha>0$ and rate $\beta>0$; $f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1}e^{-\beta x}$ for $x>0$
$X \sim \Gamma(\alpha,\theta)$	Gamma distribution with shape $\alpha>0$ and scale $\theta>0$; $f_X(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1}e^{-x/\theta}$ for $x>0$, with $\theta = 1/\beta$
$X \sim \mathrm{Beta}(\alpha,\beta)$	Beta distribution with shape parameters $\alpha,\beta>0$ and density $f_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$ for $0<x<1$
$X \sim \mathcal{N}(\mu,\sigma^2)$, $X \sim N(\mu,\sigma^2)$	Normal distribution with mean μ and variance σ^2
$\phi(x)$	Standard normal probability density function, $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$
$\Phi(x)$	Standard normal cumulative distribution function, $\Phi(x) = \int_{-\infty}^{x} \phi(t)\,\mathrm{d}t$
$X \sim \chi^2_\nu$	Chi-squared distribution with ν degrees of freedom
$X \sim t_\nu$	Student's t distribution with ν degrees of freedom
$X \sim F_{\nu_1,\nu_2}$	F distribution with numerator and denominator degrees of freedom ν_1, ν_2
$X_n \xrightarrow{\text{a.s.}} X$	Almost-sure convergence
$X_n \xrightarrow{\mathbb{P}} X$	Convergence in probability
$X_n \xrightarrow{L^p} X$	Convergence in L^p
$X \overset{\mathrm{d}}{=} Y$	Equality in distribution of the random variables X and Y
$X_n \xrightarrow{\mathrm{d}} X,\ X_n \Rightarrow X$	Convergence in distribution

9 Geometry

In triangle notation, A, B, C denote vertices and a, b, c the opposite side lengths. Lowercase letters such as ℓ, m, n may denote lines. The notation $\overrightarrow{AB}$ may denote either the ray from A through B or the vector from A to B, according to context. Greek letters such as ω, Ω, Γ may be used as local names for circles. In differential geometry, the Frenet frame follows the standard uppercase convention $\mathbf{T}, \mathbf{N}, \mathbf{B}$, typeset in bold to emphasize that these are vectors; uppercase

italic letters such as X, Y may denote vector fields in the usual field-specific convention.

Notation	Meaning
$\triangle ABC$	Triangle with vertices A, B, C
$\angle ABC$	Geometric angle with vertex B
$m\angle ABC$	Measure of the angle $\angle ABC$
$\angle(\ell, m)$	Directed angle from the line ℓ to the line m, taken modulo π, when directed angles are being used
$\overline{AB}$	Segment joining A and B
AB	Length of $\overline{AB}$
$\overrightarrow{AB}$	Ray from A through B, or vector from A to B, according to context
$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$	Vector from A to B when $\mathbf{a}$ and $\mathbf{b}$ are the corresponding position vectors
$\overleftrightarrow{AB}$	Line through A and B
$\ell \parallel m$	Lines ℓ and m are parallel
$\ell \perp m$	Lines ℓ and m are perpendicular
$\triangle ABC \cong \triangle DEF$	Congruent triangles; corresponding vertices are listed in the same order
$\triangle ABC \sim \triangle DEF$	Similar triangles; corresponding vertices are listed in the same order
$[ABC]$	Area of $\triangle ABC$
$C(O, r)$	Circle with center O and radius r
$\odot O$	Circle centered at O, especially in elementary geometry when the radius is understood
$\mathrm{Pow}(P, \omega)$	Power of P with respect to a circle ω with center O and radius r: $OP^2 - r^2$
(ABC)	Circle through the noncollinear points A, B, C; in particular, the circumcircle of $\triangle ABC$, in elementary or olympiad geometry context
$\overset{\frown}{AB}$, $\overset{\frown}{ACB}$	Arcs specified by their endpoints, with an additional point used when needed to distinguish the intended arc
O, I, H, G	Circumcenter, incenter, orthocenter, and centroid of a triangle, respectively
R, r	Circumradius and inradius of a triangle, respectively
$s = \dfrac{a+b+c}{2}$	Semiperimeter of a triangle with side lengths a, b, c
$d(P, Q)$	Distance between the points P and Q
$\mathrm{dist}(P, A)$	Distance from the point P to the set A
$\mathrm{length}(C)$, $\ell(C)$	Length of the curve C
$\mathrm{area}(S)$	Area of the surface S
$\mathrm{vol}(\Omega)$	Volume of the region Ω
$\mathrm{diam}(A)$	Diameter of the set A
$\mathrm{conv}(A)$	Convex hull of the set A
$\mathrm{aff}(A)$	Affine hull of the set A
$\mathbf{x} = \mathbf{p} + t\mathbf{v}$	Parametric equation of a line through $\mathbf{p}$ in the direction $\mathbf{v}$
$\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0$	Hyperplane through $\mathbf{p}$ with normal vector $\mathbf{n}$
$\mathbb{R}^n$	n-dimensional real Euclidean space
$S^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \|\mathbf{x}\| = 1\}$	Unit n-sphere
$S_r^{n-1}(\mathbf{a})$	Sphere of radius r centered at $\mathbf{a} \in \mathbb{R}^n$
$\mathrm{Isom}(X)$	Isometry group of the metric or geometric space X
$\mathbb{RP}^n$	Real projective n-space
$\mathbb{CP}^n$	Complex projective n-space

Notation	Meaning
$[x_0 : \cdots : x_n]$	Homogeneous coordinates in projective space
$\mathbb{H}^n$	n-dimensional hyperbolic space
$d_{\mathbb{H}}$	Hyperbolic distance, when the hyperbolic metric is understood
$\gamma\colon I \to \mathbb{R}^n$	Parametrized curve, where $I \subseteq \mathbb{R}$ is an interval
$\dot{\gamma}(t)$	Velocity or tangent vector of a differentiable parametrized curve; for a curve in a manifold M, $\dot{\gamma}(t) \in T_{\gamma(t)}M$
s	Arc-length parameter, when so specified
$\mathbf{T},\ \mathbf{N},\ \mathbf{B}$	Unit tangent, principal normal, and binormal vectors of the Frenet frame
κ	Curvature of a regular curve
τ	Torsion of a space curve
T_pM	Tangent space of the manifold or regular surface M at p
$\mathbf{n}\colon M \to S^2$	Gauss map of an oriented surface $M \subseteq \mathbb{R}^3$, assigning its unit normal vector
$\mathrm{I},\ \mathrm{II}$	First and second fundamental forms of a regular surface
$\kappa_1,\ \kappa_2$	Principal curvatures of a regular surface
$K = \kappa_1\kappa_2$	Gaussian curvature of a regular surface
$H = \dfrac{\kappa_1 + \kappa_2}{2}$	Mean curvature of a regular surface under the average convention
(M, g)	Riemannian manifold M with Riemannian metric g
g_p	Riemannian inner product on T_pM
$\nabla_X Y$	Covariant derivative of the vector field Y in the direction of the vector field X; in Riemannian geometry, ∇ commonly denotes the Levi–Civita connection
Γ_{ij}^k	Christoffel symbols of a connection in local coordinates
$\exp_p\colon \mathcal{D}_p \to M$	Riemannian exponential map at p, where $\mathcal{D}_p \subseteq T_pM$ is its maximal domain. If M is geodesically complete, then $\mathcal{D}_p = T_pM$
$\nabla_{\dot{\gamma}}\dot{\gamma} = 0$	Geodesic equation for an affinely parametrized curve γ

10 Number Theory

Notation	Meaning
$a \mid b$	a divides b.
$a \nmid b$	a does not divide b.
$\gcd(a_1, \ldots, a_n)$	Greatest common divisor
$\operatorname{lcm}(a_1, \ldots, a_n)$	Least common multiple
$p^k \parallel n$	p^k exactly divides n: $p^k \mid n$ and $p^{k+1} \nmid n$; equivalently, $v_p(n) = k$.
$v_p(n),\ \nu_p(n)$	p-adic valuation of n
$a \equiv b \pmod{n}$	a is congruent to b modulo n.
$[a]_n,\ \overline{a}$	Residue class of a modulo n; the bar is used when the modulus or quotient ring is clear
$a \bmod n$	Least nonnegative remainder of a modulo n, for $n > 0$
$\mathbb{Z}/n\mathbb{Z}$	Ring of residue classes modulo n
$(\mathbb{Z}/n\mathbb{Z})^\times$	Multiplicative group of units modulo n
$a^{-1} \pmod{n}$	Multiplicative inverse of a modulo n
$\operatorname{ord}_n(a)$	Multiplicative order of a modulo n
$\mathbb{F}_q,\ \mathrm{GF}(q)$	Finite field with q elements, where q is a prime power

Notation	Meaning
$\left(\frac{a}{p}\right)$	Legendre symbol, for an odd prime p
$\left(\frac{a}{n}\right)$	Jacobi symbol, for an odd positive integer n
$\varphi(n)$	Euler totient function
B_n	nth Bernoulli number, under the convention $B_1 = -\frac{1}{2}$
$\sum_{d\mid n}$	Sum over the positive divisors d of n
$\prod_{p\mid n}$	Product over the prime divisors p of n
$\sigma_k(n) = \sum_{d\mid n} d^k$	Sum of the kth powers of the positive divisors of n
$\tau(n) = \sigma_0(n)$	Number of positive divisors of n
$\sigma(n) = \sigma_1(n)$	Sum of positive divisors of n
$\mu(n)$	Möbius function
$\omega(n)$	Number of distinct prime divisors of n
$\Omega(n)$	Number of prime factors of n, counted with multiplicity
$\mathrm{rad}(n)$	Product of the distinct prime divisors of n
$(f * g)(n) = \sum_{d\mid n} f(d)g(n/d)$	Dirichlet convolution of arithmetic functions f and g
$\pi(x)$	Number of primes not exceeding x
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$ for a prime p and an integer $k \geq 1$, and 0 otherwise
$[a_0; a_1, a_2, \dots]$	Simple continued fraction
$[a_0; a_1, \dots, a_n] = \frac{p_n}{q_n}$	nth convergent of a continued fraction
$\Phi_n(x)$	nth cyclotomic polynomial
$\mathbb{Z}[i]$	Ring of Gaussian integers
$K = \mathbb{Q}(\sqrt{d})$	Quadratic field, with squarefree $d \in \mathbb{Z}$ and $d \neq 1$
$\mathcal{O}_K$	Ring of integers of the number field K
$\mathrm{N}(\mathfrak{a}) = \lvert\mathcal{O}_K/\mathfrak{a}\rvert$	Absolute norm of a nonzero integral ideal $\mathfrak{a} \subseteq \mathcal{O}_K$
D_K, $\mathrm{disc}(K)$	Discriminant of the number field K
$\mathrm{Cl}(K)$	Ideal class group of the number field K
$h_K = \lvert\mathrm{Cl}(K)\rvert$	Class number of the number field K

11 Algebra

G, H usually denote groups, R, S rings, I, J ideals, M, N modules, and F, K, L fields. Rings are associative and have an identity unless otherwise stated; commutativity is not assumed. Ring homomorphisms preserve multiplicative identities. Unless otherwise stated, ideals are two-sided. The zero ring is allowed, while fields and division rings are nonzero. Modules are left modules unless otherwise stated.

Notation	Meaning
G	Group
e, 1_G	Identity element of a group
g^{-1}	Inverse of $g \in G$
$\lvert G\rvert$	Order (cardinality) of G
$\mathrm{ord}(g)$, $\lvert g\rvert$	Order of the element g
$\langle S\rangle$	Subgroup generated by $S \subseteq G$

Notation	Meaning
$\langle g \rangle$	Cyclic subgroup generated by g
$H \leq G$	H is a subgroup of G.
$H \trianglelefteq G$	H is a normal subgroup of G.
$gH,\ Hg$	Left and right cosets of H in G
$[G : H]$	Index of H in G
G/H	Set of left cosets of $H \leq G$ in G; a quotient group when $H \trianglelefteq G$
$\ker \varphi$	Kernel of the homomorphism φ
$\operatorname{im} \varphi$	Image of the homomorphism φ
$\mathrm{Aut}(G)$	Automorphism group of G
$\mathrm{Inn}(G)$	Inner automorphism group of G
$\mathrm{Out}(G)$	Outer automorphism group $\mathrm{Aut}(G)/\mathrm{Inn}(G)$
$Z(G)$	Center of G
$C_G(A),\ C_G(x)$	Centralizer in G of $A \subseteq G$ or $x \in G$
$N_G(A),\ N_G(H)$	Normalizer in G of $A \subseteq G$ or $H \leq G$
$g^G = \{hgh^{-1} \mid h \in G\}$	Conjugacy class of $g \in G$
$[x, y] \coloneqq x^{-1}y^{-1}xy$	Commutator of x and y
$[G, G],\ G'$	Commutator (derived) subgroup of G
$\mathrm{Syl}_p(G)$	Set of Sylow p-subgroups of G
n_p	Number of Sylow p-subgroups of G
$G \curvearrowright X$	Action of G on X
$g \cdot x$	Action of $g \in G$ on $x \in X$
$G \cdot x,\ \mathrm{Orb}_G(x),\ \mathrm{Orb}(x)$	Orbit of x under the action of G; the group subscript may be omitted when the action is clear
$G\backslash X,\ X/G$	Set of orbits of a left G-action on X; X/G is also used when the action convention is clear
$\mathrm{Stab}_G(x),\ G_x$	Stabilizer of x in G
$\mathrm{Fix}(g),\ X^g$	Fixed-point set $\{x \in X \mid g \cdot x = x\}$ of $g \in G$ acting on X
X^H	Common fixed-point set $\{x \in X \mid h \cdot x = x \text{ for all } h \in H\}$ of $H \leq G$
$G \times H$	Direct product of groups
$N \rtimes_\varphi H$	Semidirect product determined by $\varphi\colon H \to \mathrm{Aut}(N)$, with N the normal factor
$\langle S \mid R \rangle$	Group presentation with generators S and relations R
C_n	Cyclic group of order n
D_n	Dihedral group of symmetries of a regular n-gon; $\lvert D_n \rvert = 2n$. Some texts write D_{2n} for this group, indexing instead by its order
$\mathrm{Sym}(X)$	Symmetric group of all permutations of the set X
$S_n,\ \mathfrak{S}_n$	Symmetric group on n symbols
$(a_1\, a_2\, \ldots\, a_k)$	Cycle notation for a permutation sending a_i to a_{i+1} for $i < k$ and a_k to a_1
$A_n,\ \mathfrak{A}_n$	Alternating group on n symbols
V_4	Klein four-group
Q_8	Quaternion group of order 8
$\mathrm{GL}_n(F)$	General linear group over F
$\mathrm{SL}_n(F)$	Special linear group over F
$\mathrm{O}(n)$	Orthogonal group of $n \times n$ real matrices
$\mathrm{SO}(n)$	Special orthogonal group; matrices in $\mathrm{O}(n)$ with determinant 1
$\mathrm{U}(n)$	Unitary group of $n \times n$ complex matrices
$\mathrm{SU}(n)$	Special unitary group; matrices in $\mathrm{U}(n)$ with determinant 1

Notation	Meaning
R	Ring
$0_R,\ 1_R$	Additive and multiplicative identities of R, when disambiguation is useful
$R^\times$	Group of units of R
$I,\ J$	Two-sided ideals of R, unless otherwise stated
$\mathfrak{p},\ \mathfrak{q}$	Prime ideals
$\mathrm{Spec}(R)$	Prime spectrum of a commutative ring R: the set of its prime ideals
$\mathfrak{m},\ \mathfrak{n}$	Maximal ideals
$\sqrt{I}$	Radical of the ideal I in a commutative ring R: $\{r \in R \mid r^n \in I \text{ for some } n \geq 1\}$
(a)	Two-sided ideal generated by a; in a commutative ring, the usual principal ideal
$(a_1, \ldots, a_n)$	Two-sided ideal generated by $a_1, \ldots, a_n$
$I + J$	Sum of the ideals I and J
IJ	Product of the ideals I and J
R/I	Quotient ring
$R[x],\ R[x_1, \ldots, x_n]$	Polynomial rings over R in commuting central indeterminates; the indeterminates commute with the coefficients and with one another. Noncommutative polynomial constructions are specified separately
$\deg f$	Degree of the polynomial f
$M_n(R)$	Ring of $n \times n$ matrices over R
$R[G]$	Group ring of G over R
$\mathrm{Frac}(R)$	Fraction field of the integral domain R
$\mathrm{char}\, R$	Characteristic of R
$\mathbb{H}$	Quaternion division algebra over $\mathbb{R}$
$M,\ N$	Left R-modules, unless otherwise stated
${}_RM,\ M_R$	Left and right R-modules, when the side is emphasized
M/N	Quotient module, for a submodule N of M
$M \oplus N$	Direct sum of modules
$\bigoplus_{i \in \Lambda} M_i$	Direct sum of an indexed family of modules, with index set Λ
R^n	Free R-module of rank n
$\mathrm{Hom}_R(M, N)$	Set of R-module homomorphisms from M to N
$\mathrm{End}_R(M)$	Endomorphism ring of the R-module M
$\mathrm{Aut}_R(M)$	Automorphism group of the R-module M
$\mathrm{Ann}_R(M)$	Annihilator of the R-module M
$M_R \otimes_R {}_RN,\ M \otimes_R N$	Tensor product over R; the first notation emphasizes that M is a right R-module and N a left R-module. The side distinction is normally suppressed when R is commutative or the required module structures are otherwise clear
K/F	Field extension of F by K
$[K : F]$	Degree of the field extension K/F
$F(\alpha),\ F(\alpha_1, \ldots, \alpha_n)$	Fields generated over F by the indicated elements
$F[\alpha]$	Subring (equivalently, F-subalgebra) generated by F and α
$m_{\alpha,F}(x)$	Minimal polynomial of α over F
$\overline{F}$	Algebraic closure of F
$\mathrm{Gal}(L/K)$	Galois group of L/K
L^H	Fixed field of $H \leq \mathrm{Gal}(L/K)$

Notation	Meaning
$\mathrm{Tr}_{L/K}(\alpha)$	Field trace of $\alpha \in L$ from L to K
$\mathrm{N}_{L/K}(\alpha)$	Field norm of $\alpha \in L$ from L to K
$\mathrm{trdeg}_K L$	Transcendence degree of L over K

12 Topology

Notation	Meaning
$(X, \mathcal{T})$	Topological space X with topology $\mathcal{T}$
$\mathcal{T}$	Topology on the underlying set X, when X is understood
$\mathcal{B}$	Basis for a topology
$\mathcal{S}$	Subbasis for a topology
$A^\circ,\ \mathrm{int}\, A$	Interior of A
$\overline{A},\ \mathrm{cl}\, A$	Closure of A
∂A	Boundary of A
$\mathrm{ext}\, A$	Exterior of A in X: $\mathrm{int}(X \setminus A) = X \setminus \overline{A}$
A'	Derived set of A: the set of its accumulation points in X
(X, d)	Metric space X with metric d
$\mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}\}$	Subspace topology on $A \subseteq X$
$\pi_i \colon \prod_{j \in J} X_j \to X_i$	Coordinate projection of a product space, for $i \in J$
$X/\!\sim$	Quotient space when the quotient set is equipped with the quotient topology
$q \colon X \twoheadrightarrow X/\!\sim$	Canonical quotient map
X/A	Quotient space obtained by collapsing the subspace $A \subseteq X$ to a point
$C(X, Y)$	Set of continuous maps from X to Y
$X \cong Y$	Homeomorphic spaces
$\mathrm{Homeo}(X)$	Group of self-homeomorphisms of X
$\iota \colon A \hookrightarrow X$	Inclusion map of a subspace $A \subseteq X$
$f \colon X \hookrightarrow Y$	Topological embedding when explicitly stated; the hook arrow by itself retains its general meaning of injectivity
(X, x_0)	Pointed space with basepoint x_0
$I = [0, 1]$	Unit interval, especially for paths and homotopies
$\gamma \colon I \to X$	Path in X
$\gamma(0) = \gamma(1) = x_0$	Loop based at x_0
$H \colon X \times I \to Y$	Homotopy between maps $X \to Y$
$f \simeq g$	Homotopic maps $f, g \colon X \to Y$
$X \simeq Y$	Homotopy equivalent spaces
$[\gamma]$	Homotopy class of a path or loop γ, according to context
$\pi_0(X)$	Set of path-connected components of X; not generally a group
$\pi_1(X, x_0)$	Fundamental group of X based at x_0
$\pi_n(X, x_0)$	nth homotopy group of the pointed space (X, x_0), for $n \geq 1$
f_*	Homomorphism induced by a map f on a covariant invariant such as homotopy or homology, with the invariant clear from context
f^*	Homomorphism induced by a map f on a contravariant invariant such as cohomology, with the invariant clear from context
$p \colon \widetilde{X} \to X$	Covering map, with $\widetilde{X}$ a covering space of X

Notation	Meaning
$\widetilde{X}$	Covering space; it denotes a universal cover only when explicitly specified
$\mathrm{Deck}(p)$	Group of deck transformations of the covering p
$D^n = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq 1\}$	Closed n-disk
$X \vee Y$	Wedge sum of pointed spaces X and Y
T_0, T_1, T_2, T_3, T_4	Separation axioms; here T_2 means Hausdorff, T_3 means T_1 and regular, and T_4 means T_1 and normal
M^n	n-dimensional topological manifold, when the dimension is displayed
∂M	Boundary of a manifold M
Σ_g	Closed connected orientable surface of genus g
$M \# N$	Connected sum of manifolds M and N
$\chi(X)$	Euler characteristic of X
$H_n(X; G)$	nth homology group of X with coefficients in G
$H_n(X)$	Integral homology group $H_n(X; \mathbb{Z})$ when coefficients are omitted
$\widetilde{H}_n(X; G)$	nth reduced homology group of X with coefficients in G
$H^n(X; G)$	nth cohomology group of X with coefficients in G
$b_n(X)$	nth Betti number; for example, $b_n(X) = \dim_{\mathbb{Q}} H_n(X; \mathbb{Q})$ when this dimension is finite

13 Miscellaneous Notation

Notation	Meaning
$\lfloor x \rfloor$	Floor of x
$\lceil x \rceil$	Ceiling of x
$\{x\}$	Fractional part $x - \lfloor x \rfloor$, when unambiguous
$\mathrm{sgn}(x)$	Sign of a real number or permutation, according to context
$a \propto b$	a is proportional to b; in a functional context, the proportionality factor is independent of the variable under discussion.
$O(g(n))$	Big-O upper asymptotic bound
$o(g(n))$	Little-o strict asymptotic bound
$\Theta(g(n))$	Two-sided asymptotic order
$\Omega(g(n))$	Asymptotic lower bound
$f \lesssim g,\ f \gtrsim g$	Upper and lower bounds up to a positive constant factor, respectively, in a stated regime; for nonnegative functions, $f \lesssim g$ means $f(x) \leq C g(x)$ for some constant $C > 0$
$f(x) \asymp g(x)$	Comparable up to positive multiplicative constants in a stated regime: $c\, g(x) \leq f(x) \leq C\, g(x)$ for some $0 < c \leq C$ and nonnegative g
$\sum_{\mathrm{cyc}} f(a, b, c)$	Cyclic sum over the cyclic permutations of the displayed variables; for three variables, $f(a, b, c) + f(b, c, a) + f(c, a, b)$
$\sum_{\mathrm{sym}} f(a, b, c)$	Symmetric sum over the distinct permutations of the displayed variables in the summand; the variables and any normalization are stated when needed
Δq	Finite change in a quantity q, especially in a discrete or applied context
δ_{ij}	Kronecker delta: 1 if $i = j$ and 0 otherwise

ϵ_{ijk}	Three-dimensional Levi–Civita symbol: the sign of the permutation (i, j, k) when (i, j, k) is a permutation of $(1, 2, 3)$, and 0 otherwise